



Determinants of static balance performance across lifespan vary with sensory conditions

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Abstract

Age-related changes in balance control generally describes a U-shaped relation, with lesser stability observed at both ends of the lifespan. However, the determinants that drive this relation are still not clear and may vary with the sensory conditions in which the postural task is performed. This cross-sectional study investigated age-related changes in static balance control from childhood to old age and investigated potential factors underlying balance performance. One hundred fifty-seven participants were divided into six age groups (6–14; 14–17; 18–39; 40–59; 60–75; >76 yrs). Mean velocity (MV) of the centre of pressure was recorded while participants stood upright under four sensory conditions: eyes open or closed on stable (top plate of the force platform) and unstable (foam mat) surface. Surface electromyography of the soleus, gastrocnemius medialis, and tibialis anterior was recorded during postural tasks. Ankle joint position sense (aJPS) and maximal isometric force of plantar flexors and dorsiflexors were also measured. Group differences in MV were observed across all conditions ($p < 0.001$), with greater MV in children (6–14 yrs) and older adults (>60 yrs), especially on the unstable surface for older adults. Regression analyses showed that greater aJPS acuity contributes to stabilize CoP on rigid support, whereas an increase in leg muscle activity may help compensate for the inherent instability induced by the compliant surface. Chronological age appears as an independent predictor of static balance control only in this latter condition. These results highlight the necessity of using various sensory conditions to assess balance performance and its underlying determinants across lifespan.

Keywords Balance · Postural sway · Ageing · Proprioception · Muscle activity

Introduction

Static balance control, defined as the ability to maintain stability during upright standing position, is a fundamental requisite to perform numerous daily life activities. Balance impairment, in contrast, is a major factor for falls risk,

which can lead to injury and disability, particularly among older adults (Lord et al. 2003; Tinetti et al. 1988). Effective balance control relies on the integration of visual, vestibular, and proprioceptive information, dynamically modulated by task demands and environmental context (Lephart et al. 1997; Peterka 2002). From childhood to old age, changes in the processing of sensory inputs to generate appropriate motor responses for maintaining stability (sensorimotor integration) might influence the ability to control balance under various conditions (Horak 2006). Despite extensive research in specific age groups (Abrahamová and Hlavacka 2008; Brech et al. 2022; Prado et al. 2007; Roman-Liu 2018), only a few studies have examined balance control from childhood to old adulthood (Hytönen et al. 1993; Humbeeck et al. 2023). These studies indicate a U-shaped function between balance and age, with the poorest balance control observed at both ends of the lifespan. Yet, gaps still exist in our understanding of the determinants of balance

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performance across lifespan (Behm et al. 2015; Kiss et al. 2018; Woollacott and Shumway-Cook 1990).

Quiet upright standing is often considered as an inverted pendulum rotating around the ankle joint (Winter et al. 1998), with proprioceptive inputs being a major source of information (Peterka 2002). Age-related declines in proprioceptive acuity, often accompanied by an increased reliance to visual inputs (Henry and Baudry 2019), have been linked to increased postural sway and fall risk (Lord et al. 1991; Shaffer and Harrison 2007). However, when investigating the ankle joint position sense (aJPS) – the ability to perceive angular ankle position – no age-related changes were observed across lifespan (Theodosiadou et al. 2025), questioning the influence of altered proprioception on balance control, which may depend on support surface (Kiers et al. 2012). In contrast, increased coactivation of the agonist and antagonist leg muscles is a common observation in older adults during upright standing and may reflect compensatory strategies to stabilize posture (DeVita and Hortobagyi 2000; Laughton et al. 2003). While these mentioned factors could influence the ability to maintain stable upright standing, their contribution in balance performance across lifespan remains poorly documented.

Furthermore, as the capacity to maintain balance relies on a complex interplay between sensory inputs and motor command (Horak 2006), the sensory conditions of the postural task can influence balance performance. Standing on foam surface increases instability, as assessed by an increase in centre of pressure (CoP) excursion (Abrahamová and Hlavacka 2008; Baudry and Duchateau 2012). This could reflect, among other factors, an increased complexity to interpret proprioceptive signals and difficulties to apply adequate foot pressure on the compliant surface (Kiers et al. 2012), which can be worsened when standing with eyes closed.

This study aimed to investigate age-related changes in static balance control, from childhood to old age, by assessing CoP mean velocity (MV) during upright standing performed in different sensory conditions. Furthermore, we seek to identify determinants of balance performance in each sensory condition. These findings may provide a basis for future research aimed at developing targeted interventions to prevent falls and balance impairments.

Methods

Participants

One-hundred fifty-seven individuals aged from 6 to 92 yrs participated in this multicentre study (Table 1). Children were divided in two groups based on their maturation offset (Mirwald et al. 2002): Young-Children (YC: 6–14 yrs); Older-Children (OC: 14–17 yrs). Even though large anthropometric changes occur in young ages, we first accounted for maturation status given the well-established influence of biological maturity on neuromuscular performance (Fernandez-Fernandez et al. 2022; Sinkovic et al. 2023). Furthermore, body mass and height were included as covariates in the regression analyses to control for anthropometric differences in the younger group. Adults were divided into four groups: Young (Y: 18–39 yrs), Middle-aged (M: 40–59 yrs), Young-Old (YO: 60–75 yrs) and Old-Old (OO: >76 yrs). Approval for the project was obtained from the local Ethics Committee of the Université libre de Bruxelles (B4062021000206) and of the Aristotle University of Thessaloniki (25284/2022). All procedures were in accordance with the ethical standards of Declaration of Helsinki. An informed written consent was signed by all participants or minors' guardians before starting the experimental procedures, and minor participants provided their own assent to participate to the study.

Individuals with neurological diseases such as Parkinson's, multiple sclerosis, or stroke, as well as volunteers with orthopaedic problems within 12 months prior to the study were excluded. None of the participants were taking medications that could influence balance. For participants over 60 years old, two tests were performed to assess their cognitive (*Montreal Cognitive Assessment*; MoCA) and psychological state (*Geriatric Depression Scale*; GDS). Participants with inconclusive scores (MoCA > 26 or GDS < 10) were excluded from the study. Each participant completed a short questionnaire to measure the habitual physical activity (*International Physical Activity Questionnaire*; IPAQ).

Table 1 Anthropometric data of the participants. Data are presented as mean (SD)

Group	YC (n=15)	OC (n=15)	Y (n=32)	M (n=40)	O (n=24)	OO (n=31)
Age (yrs)	10.3 (2.5)	15.9 (1.0)	29.1 (5.6)	49.8 (5.5)	68.7 (4.9)	80.3 (3.7)
Height (cm)	154 (10.7)	168 (8.0)	172 (9.3)	170 (8.4)	166 (9.5)	167 (10.1)
Mass (kg)	45.7 (13.1)	65.6 (12.6)	69.8 (12.9)	73.5 (11.3)	68.3 (11.0)	69.3 (10.9)
IPAQ (/3)	2.7 (0.4)	2.8 (0.5)	2.5 (0.7)	2.3 (0.6)	2.4 (0.7)	2.2 (0.7)

YC young-Children (6–14 yrs), OC Old-Children (14–17 yrs), Y young (18–39 yrs), M Middle-aged (40–59 yrs), YO young-Old (60–75 yrs), OO Old-Old (76–92 yrs). IPAQ International Physical Activity Questionnaire (from 1 to 3)

Materials

Surface electromyogram

Electromyography (EMG) was assessed by placing auto-adhesive electrodes in a bipolar configuration with a centre-to-centre distance of 2 cm to record soleus (SOL), gastrocnemius medialis (GM) and tibialis anterior (TA) of the dominant leg. The skin was previously shaved when necessary and cleaned with alcohol to reduce the impedance at the skin–electrode interface. The electrodes for SOL were placed 2 cm below the muscle–tendon junction of the GM in line with the Achilles tendon and for the GM were placed midway between the femoral condyle and the muscle–tendon junction. For the TA they were placed at one third of the distance between the fibular head and the lateral malleolus, and 1 cm lateral to the tibia. Finally, the three reference electrodes (one for each muscle) were placed over the anterior border of the tibia bone. The EMG signals were amplified (1000×) and band-pass filtered (10–1000 Hz) prior to A/D sampling (2 kHz; Power 1401, 16-bit resolution, Cambridge Electronic Design, UK for *Bruxelles* and Biopac System Inc., Goleta, CA, USA for *Serres*).

Force platform

A force platform (AMTI, OR6-6- 2000, Advanced Mechanical Technology, Watertown, MA, USA for *Bruxelles* and 9281CA Kistler Instruments Ltd., Winterthur, Switzerland for *Serres*) was used to record the CoP mean velocity (MV) (Power 1401, 16-bit resolution, Cambridge Electronic Design, UK, 100 Hz for *Bruxelles* and 1000 Hz, BioWare, Version 3.21, Kistler for *Serres*).

Dynamometer

To record plantar flexors (PF) and dorsiflexors (DF) force, participants were seated in a custom-made dynamometer. The ankle and knee angles of the dominant leg were positioned at 90° and 0° (full extension), respectively, and the foot tightly attached to a footplate connected to a force transducer (model 4576A2N, Kistler, Winterthur, Switzerland; 200 Hz for *Bruxelles* and CCT transducers TF 022, OT Bioelettronica, Turin, Italy; 1000 Hz for *Serres*).

Experimental procedure

Balance control

Participants stood upright barefoot on the force platform, arms at their sides and head looking straight, in a self-selected comfortable feet position, which was marked on a

paper to replace the feet in the same position throughout the different balance conditions. Participants were asked to stay as steady as possible. A red point was fixed 1.5 m in front of them at eyes level to avoid head movements. Four different conditions were performed in a random order: eyes open and closed on a stable (EOS and ECS) and unstable (foam pad; 48 × 40 × 6 cm, Airex, Sins, Switzerland) surface (EOU and ECU). In ECS and ECU, a sleeping mask covered the eyes. Participants repeated 3 trials of 60 s in each condition with at least 60 s rest between trials.

Proprioceptive acuity

Proprioceptive acuity was assessed with the aJPS of the dominant limb during an ipsilateral reproduction (IRT) and a contralateral matching task (CMT). Detailed descriptions of this assessment can be found in Theodosiadou et al. (2025). Briefly, the participants were seated in an adaptable armchair with their ankles and knees flexed at 90°. The feet were placed on two adjustable pedals of a wooden custom-made construction, with a wireless inertial measurement unit (BWT901CL, WitMotion, Shenzhen Co., Ltd) fixed on the pedal of the dominant foot to provide a proxy of the ankle angle. The IRT consisted of reproducing a previously experienced ankle angle with the same limb while the CMT consisted of matching the passively held target ankle angle of one limb with the contralateral ankle. Both tests were performed blind-folded from 5° (DF10) and 20° (DF25) plantarflexion position to a 5° dorsiflexion target position. The magnitude of the error (expressed in absolute values) between the target and the reproduction position was considered for further analysis.

Maximal voluntary contraction

To record maximal voluntary contractions (MVC), participants were seated on the dynamometer (described above) and performed at least 3 MVCs of PF and DF, lasting 3–4 s, in a random order with at least 120 s rest between trials. A custom-made script (0.5-s moving window) was used to find the greatest value during the MVC force. If the difference between the best two trials was more than a 5%, additional trials were performed. All participants fulfilled 5% criterion within 5 or less MVCs trials.

Data reduction

For balance control, a 40-s epoch window (excluding the first 10 s considered as a stabilization phase) was cut off (Baudry et al. 2015) during which CoP MV (5-Hz low pass filtered), which was the CoP path length divided by the time (mm/s), as well as the rectified mean EMG of the SOL, GM

and TA during postural tasks was analysed with custom-made MATLAB (MATLAB-R2022a 64-bit, The Mathworks Inc., Natick MA, USA) script. The EMG associated with peak torque during MVC (EMG_{MVC}) was determined over a 1-s epoch (500ms before and after the peak torque value) and was used as the reference to normalize the EMG during balance tasks. For both MV and EMG, the average value of the 3 trials was used for analysis.

Statistical analysis

Statistical analysis was performed with *JASP* software (Version 0.19.2.0). The Gaussian distribution of the data was confirmed by the Shapiro-Wilks normality test. Group effect was tested by one-way analysis of variance (ANOVA). Tukey post-hoc tests were performed in case of significant ANOVA. The best fit between MV and age was explored by comparing the goodness of fit between first- and second-degree polynomials for each balance condition. Forward stepwise linear multiple regressions were performed to identify the variables that contribute to static balance control across lifespan in each sensory condition. The dependent variable was MV and predictors were age, body mass, height, SOL, GM and TA ($\%EMG_{MVC}$), PF and DF force (expressed relative to body mass), aJPS (IRT_{DF10} , IRT_{DF25} , CMT_{DF10} , CMT_{DF25}) and IPAQ score. However, to properly capture second-order effect of age on MV while retaining the linear term needed for full interpretation of the trend, “age” was included as age squared (age^2) in the multiple regression. The statistical significance level was set to 0.05. Eta-squared (η^2) was used as an estimate of effect size.

Results

Effect of age on static balance control

The one-way ANOVA showed group differences in static balance control for all conditions ($p < 0.001$, EOS: $F_{5,151}=6.8$, $\eta^2=0.18$; ECS: $F_{5,151}=5.4$, $\eta^2=0.15$; EOU: $F_{5,151}=16.1$, $\eta^2=0.35$; ECU: $F_{5,151}=16.2$, $\eta^2=0.35$). Post-hoc

comparisons showed that children (YC and OC) and older adults (YO and OO) exhibited greater MV compared to Y and M in stable conditions. In unstable conditions, YO and OO presented greater MV than YC, OC, Y and M (Table 2).

In all sensory conditions, a second-order polynomial function better fitted the relation between MV and age (Fig. 1) compared with a first-order linear function (EOS: $r < 0.001$; ECS: $r = 0.003$; EOU: $r = 0.24$; ECU: $r = 0.28$, $p < 0.001$) with greater correlation coefficient, although it remains small, for unstable (EOU: $r = 0.40$, $p < 0.001$; ECU: $r = 0.40$, $p < 0.001$) than stable surface (EOS: $r = 0.20$, $p < 0.001$; ECS: $r = 0.16$, $p < 0.001$).

Factors contributing to MV

Multiple regression analyses identified specific sets of predictors for each sensory condition (Table 3). The part of the MV variance explained by the model increased with the increasing difficulty of the task, ranging for adjusted R^2 from 0.05 to 0.46. aJPS was positively associated with MV on stable surface. In contrast, on unstable surface, muscle activity and age were significant predictors. For ECU height also contributed to MV variance.

Values of all variables included to the multiple analyses can be found in Table 4. More details about statistical group differences for these variables are presented in supplementary materials.

Discussion

The results of this study indicate that children and older adults exhibit greater postural sway than young and middle-aged adults during upright standing. However, older individuals have greater MV in the more complex situations than the other age groups, highlighting the decreased capacity to maintain stability with advancing age when proprioceptive and visual inputs are restricted (Peterka 2002; Horak 2006; Papavasileiou et al. 2022). Interestingly, aJPS appears as a predictor of static balance control on the stable

Table 2 Mean velocity of the centre of pressure during upright standing on stable surface with eyes open (MV_{EOS}) and closed (MV_{ECS}), and when standing on unstable surface with eyes open (MV_{EOU}) and closed (MV_{ECU})

Variables	YC	OC	Y	M	YO	OO
MV_{EOS} ($mm \cdot s^{-1}$)	10.3 (2.8)	8.5 (3.1)	6.2 (1.5)*†	6.5 (1.9)*†	7.5 (2.6)†	8.9 (4.6)
MV_{ECS} ($mm \cdot s^{-1}$)	12.7 (3.7)	10.0 (3.2)	7.7 (1.9)*†	9.0 (3.5)†	9.1 (3.7)	11.6 (5.8)
MV_{EOU} ($mm \cdot s^{-1}$)	14.8 (2.6)*	12.6 (2.3)*#	11.8 (2.6)*#	13.4 (4.2)*#	17.7 (5.6)	21.4 (7.9)
MV_{ECU} ($mm \cdot s^{-1}$)	22.3 (3.0)*	22.2 (7.7)*	21.0 (5.1)*#	24.2 (9.0)*	31.2 (19.4)*	43.3 (19.4)

Data are presented as mean (SD)

MV Mean velocity of the centre of pressure, *YC* Young-Children (6–14 yrs), *OC* Old-Children (14–17 yrs), *Y* Young (18–39 yrs), *M* Middle-aged (40–59 yrs), *YO* Young-Old (60–75 yrs), *OO* Old-Old (76–92 yrs)

Statistically significant difference ($p < 0.05$) compared to YC (†), YO (#) and OO (*)

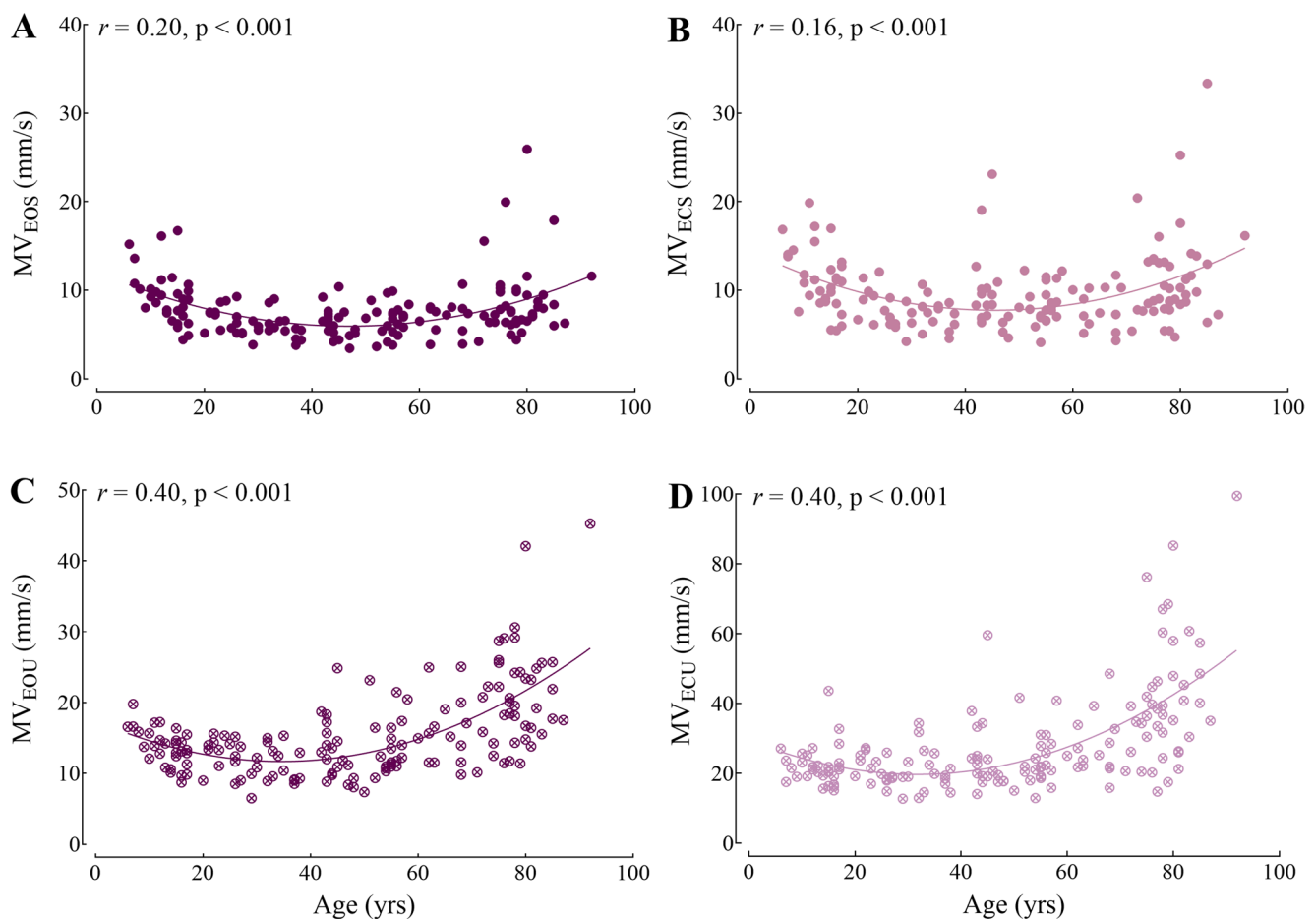


Fig. 1 Simple second order polynomial regressions between age and the mean velocity of the centre of pressure (MV) in the following conditions: on stable surface with eyes open (EOS) and closed (ECS),

and on unstable surface with eyes open (EOU), and closed (ECU). A U-shape trend was observed in stable conditions, while older adults seem to produce greater postural oscillations in unstable conditions

surface, while age and leg muscle activity partially explain balance performance on the unstable surface.

Non-linear changes in static balance control across the lifespan

This study confirmed the U-shaped relation between MV and age, which indicates that balance control initially improves from childhood to adulthood and later deteriorates with ageing (Humbeeck et al. 2023). This U-shaped function in the four conditions highlights a genuine change in balance control across lifespan, regardless of sensory condition (Humbeeck et al. 2023; Woollacott and Shumway-Cook 1990). This likely reflects, for the descending limb of the relation, the development of proprioceptive acuity and muscle strength during childhood, along with improved sensory integration (Woollacott and Shumway-Cook 1990; Hirabayashi and Iwasaki 1995) and progressive brain maturation (Koolschijn and Crone 2013; Lenroot and Giedd 2006), leading to better balance control (Roncesvalles et al.

2001; Steindl et al. 2006). Furthermore, better attentional capabilities (focus during postural tasks) may contribute to improved balance performances from children to young adults (Baccouch et al. 2015). In contrast, the ascending limb of the relation is probably due to age-related declines in sensory function (e.g., reduced proprioceptive and visual acuity), muscle strength, and neural processing effectiveness, which likely impair the ability to maintain balance stability (Henry and Baudry 2019; Lord et al. 1991; Laughton et al. 2003). In addition, this relation was strengthened for unstable surfaces, indicating that the effect of age on balance performance strongly depend on the sensory condition of the postural task. This is consistent with previous work, focusing on adulthood, reporting a non-linear increase in MV with age, more so in EOU and ECU, when the demands for sensory integration and neuromuscular control were higher (Abrahamová and Hlavacka 2008).

Table 3 Results of the multiple regression analysis performed to identify factors contributing to the change in the mean velocity of the centre of pressure with age during upright standing on stable surface with eyes open (MV_{EOS}) and closed (MV_{ECS}), and when standing on unstable surface with eyes open (MV_{EOU}) and closed (MV_{ECU})

Variable	R ² adjusted	β (SE)	t	p-value	Partial r
MV _{EOS}					
Constant	0.05	6.21 (0.52)	11.90	<0.001	
IRT _{DF25}		0.23 (0.10)	2.25	0.026	0.18
CMT _{DF25}		0.16 (0.08)	2.08	0.039	0.17
MV _{ECS}					
Constant	0.10	7.13 (0.69)	10.30	<0.001	
IRT _{DF25}		0.40 (0.13)	2.99	0.003	0.24
SOL _{ECS}		0.07 (0.03)	2.64	0.009	0.21
MV _{EOU}					
Constant	0.33	10.77 (0.70)	15.38	<0.001	
Age		0.001 (0.001)	6.73	<0.001	0.48
GM _{EOU}		0.08 (0.04)	2.10	0.037	0.17
MV _{ECU}					
Constant	0.46	-24.32 (13.61)	-1.79	0.076	
Age		0.008 (0.002)	4.11	<0.001	0.23
Height		0.23 (0.08)	2.88	0.005	0.23
TA _{ECU}		0.58 (0.13)	4.56	<0.001	0.35
SOL _{ECU}		0.23 (0.07)	3.28	0.001	0.19

The predictors were age (²), body mass (kg), height (cm), EMG of SOL, GM and TA (%EMG_{MVC}) as well as the maximal force of PF and DF (expressed relative to the body mass) and the ankle joint position sense [assessed with ipsilateral reproduction task (IRT_{DF25} and IRT_{DF10}) and contralateral matching task (CMT_{DF25} and CMT_{DF10})], and the level of physical activity measured by the IPAQ

β : regression coefficient (unstandardized), SE Standard error

Role of proprioceptive acuity and muscle activity in balance control

The multiple regression analyses revealed that aJPS (proprioception's modality) was a weak but significant predictor of balance performance on stable surface. This aligns with the idea that proprioceptive signals are more reliable and accurate when standing on stable surface (Kiers et al. 2012), making them a critical source of sensory input to maintain balance (Lord et al. 1991; Fitzpatrick and McCloskey 1994). When standing on a rigid floor, α - γ coactivation plays a crucial role in signalling a forward body tilt (Burke and Eklund 1977). This mechanism links increased activity of the plantar flexors to an enhanced proprioceptive signal, allowing muscle afferents to reliably indicate forward tilt (Tokuno et al. 1985; Johannsson et al. 2015).

However, on compliant support, active adjustments, with larger ankle movements compared to rigid surface (Hessel et al. 2021), are required to counteract the flawed feet-support interaction. These stabilizing reactions diminish the relevance of proprioceptive inputs from leg muscles as muscle activity and thereby proprioceptive signal is not linked to unequivocal direction of the CoP. Additionally, as

the foam compresses, the pressure exerted by the ground is redistributed in a non-linear manner, meaning that the force applied by the feet does not directly correspond to the body's displacement (Gosselin and Fagan 2015), leading to altered somatosensory feedback. This instability likely requires greater muscle activity compared to a rigid surface to counteract larger body sways (Sozzi et al. 2013). Furthermore, Billot and colleagues (Billot et al. 2010) reported a positive relation between postural sway and leg muscle activity, such that individuals exhibiting greater EMG activity were less stable. In the context of ageing process, where muscle strength generally decreases (Hunter et al. 2016), the increase in EMG may also represent a greater relative force compared with young adults to maintain balance. In agreement, leg muscle activity and age appear as strong predictors of balance performance when standing on foam, more so with eyes closed.

The significant contribution of age (positive partial correlation) further underscores the impact of age-related declines in neuromuscular function on balance during complex postural tasks, which likely stems from the cumulative effect of declines in various systems, including the vestibular system. A study assessing postural sway on foam surface found increased anteroposterior acceleration of the head's centre of mass in older compared to younger individuals (Takeuchi et al. 2020). This greater head acceleration should increase inputs from the vestibular system. However, age-related alterations of the vestibular system (Shaffer and Harrison 2007) may limit the vestibular inputs' integration, altering thereby the sensorimotor integration process (Dieuleveult et al. 2017) and balance, especially under distorted sensory input conditions. Nonetheless, compared to other experimental conditions using much higher unstable balance surfaces than ours, the head rotation when standing on a foam mat was limited to 2.2 degrees (Bogaart et al. 2022) indicating therefore that the contribution of the vestibular inputs should remain modest in our experimental conditions (Diaz-Artiles and Karmali 2021).

Finally, height was a significant predictor of static balance performance in the most challenging condition. This finding can be explained by the biomechanics of the inverted pendulum model, where taller individuals have a longer lever arm, resulting in greater displacement of the centre of mass during the postural sway (Winter et al. 1998). The larger sway during ECU condition likely amplifies this effect, accentuating height's contribution in this context. Accordingly, height has been shown to influence balance control, with taller individuals experiencing greater displacement of the centre of mass when standing on unstable surface (Alonso et al. 2015; Kejonen et al. 2003).

Table 4 Values of variables included in the multiple regression. young (YC) and Old-Children (OC), young (Y) and Middle-aged adults (M), and in young-Old (YO), and Old-Old adults (OO) during upright standing on stable surface with eyes open (EOS) and closed (ECS), and when standing on unstable surface with eyes open (EOU) and closed (ECU)

Variables	YC	OC	Y	M	YO	OO
Muscle activity						
SOL _{EOS} (%SOL _{max})	15.6 (11.3)	12.7 (7.4)	9.14 (6.3)	17.0 (10.3)	18.0 (12.8)	25.3 (16)
SOL _{ECS} (%SOL _{max})	15.8 (10.4)	15.0 (8.2)	10.2 (6.2)	18.8 (11.2)	18.1 (12)	25.1 (14.5)
SOL _{EOU} (%SOL _{max})	16.3 (10.3)	14.5 (10.2)	10.7 (6.1)	20.4 (10.5)	21.1 (11.4)	31.0 (15.8)
SOL _{ECU} (%SOL _{max})	17.6 (9.7)	16.0 (8.7)	12.3 (7.6)	19.9 (10.4)	22.1 (11.3)	34.0 (17.4)
GM _{EOS} (%GM _{max})	4.6 (4.2)	4.0 (3.3)	5.86 (5.2)	7.0 (4.7)	11.2 (10.0)	14.2 (7.7)
GM _{ECS} (%GM _{max})	5.9 (5.0)	7.3 (7.5)	7.9 (8.7)	8.2 (4.7)	12.5 (8.8)	14.4 (7.6)
GM _{EOU} (%GM _{max})	6.7 (4.3)	8.6 (7.4)	9.9 (10.2)	12.3 (8.8)	18.2 (12.5)	20.0 (12.9)
GM _{ECU} (%GM _{max})	10.9 (12.5)	10.6 (9.0)	14.6 (18.2)	14.2 (8.9)	23.2 (18.9)	25.3 (11.9)
TA _{EOS} (%TA _{max})	3.1 (3.7)	2.8 (2.3)	2.5 (4.1)	2.5 (2.7)	3.7 (2.6)	6.6 (11.0)
TA _{ECS} (%TA _{max})	3.0 (2.9)	2.4 (1.3)	2.1 (3.0)	2.5 (2.5)	4.0 (2.8)	6.7 (8.9)
TA _{EOU} (%TA _{max})	2.5 (1.1)	3.2 (2.2)	2.7 (2.7)	3.9 (4.2)	6.6 (4.9)	9.0 (6.9)
TA _{ECU} (%TA _{max})	3.7 (2.6)	3.7 (2.0)	3.8 (4.2)	5.3 (4.8)	10.1 (6.8)	18.3 (11.7)
Maximal force						
MVC _{PF} (body mass)	3.0 (0.8)	2.9 (0.7)	2.9 (1.0)	2.5 (0.8)	1.7 (0.6)	1.4 (0.5)
MVC _{DF} (body mass)	0.8 (0.2)	0.9 (0.3)	0.8 (0.3)	0.7 (0.2)	0.5 (0.2)	0.4 (0.1)
Ankle joint position sense						
IRT _{DF25} (°)	5.6 (5.7)	3.6 (1.1)	2.5 (1.2)	3.0 (1.9)	3.5 (1.7)	3.1 (1.8)
IRT _{DF10} (°)	2.6 (1.7)	1.7 (0.8)	1.8 (1.3)	2.0 (1.5)	2.4 (1.4)	1.8 (1.0)
CMT _{DF25} (°)	6.5 (4.2)	3.4 (3.1)	3.7 (3.8)	3.8 (2.5)	4.3 (3.2)	4.8 (3.0)
CMT _{DF10} (°)	5.3 (4.0)	3.7 (2.4)	3.5 (2.4)	3.4 (2.2)	4.5 (2.6)	4.1 (2.3)

Values are expressed as the mean (SD)

SOL soleus, GM gastrocnemius medialis, TA tibialis anterior, MVC maximal voluntary isometric contraction, PF plantarflexors, DF dorsiflexors, IRT ipsilateral reproduction task, CMT contralateral matching task

°Angle degrees

Limitations and perspectives

While this study, which used a large sample, provides valuable insights into age-related changes in balance control, some methodological considerations should be acknowledged. Our study investigated potential contributors to the age-related decline in balance, but it did not assess other determinants, such as vestibular function, ankle mobility or pain, which may also contribute to postural stability. However, none of the participants reported pain and ankle range of motion did not differ between groups, as observed when assessing ankle proprioception (Theodosiadou et al. 2025). Regarding the muscle activity, in this study only ankle joint muscles were assessed. However, hip muscles may also give relevant information during standing position (Davis et al. 2020). Finally, different compensation strategies (knee, hip etc.) have not been controlled.

On the other side, our findings can suggest that foams could be used to train muscle strength control by the repetitive contractions required to maintain upright standing. However, this could mainly be relevant for weaker individuals. Foam conditions could also force individuals to rely on sensory sources other than proprioception, which could be particularly beneficial for older adults who can present

proprioceptive deficits. Future research should explore these factors and investigate the effectiveness of targeted interventions to improve balance across the lifespan. Overall, these results advocate for tailored rehabilitation programs that incorporate proprioceptive training, strength exercises, and task-specific balance challenges that could help mitigate age-related declines in postural stability and reduce the risk of falls in older adults.

Conclusion

This study suggests that static balance undergoes significant age-related changes, with specific determinants varying depending on the characteristics of the task. A novel finding of this study is the relevant role of ankle position sense while standing on rigid support and the potential role of ankle muscle activity as a compensatory mechanism for maintaining stability during challenging conditions. These findings underscore the need for comprehensive balance assessments that incorporate both stable and unstable conditions to fully understand the underlying mechanisms of balance control. Overall, this study provides new insights into balance performance across lifespan, offering potential

direction for designing targeted interventions to improve stability.

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Data availability The datasets generated during and/or analysed during the current study are available in a data repository.

Declarations

Conflict of interest The authors have no conflicts of interest to declare.

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